Brian Takita

Shipping production AI systems and building agent tooling - real-time **multimodal** pipelines, **spec-driven** development infrastructure, **context engineering**. **20+ years** full-stack experience.

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SUMMARY

Architected a production real-time **multimodal** AI avatar pipeline at Presence AI - text, audio, and video lip-sync on **NVIDIA H100 GPUs** via **WebRTC**. Now building agent tooling in Rust: git-native correspondence (corky), **document-as-interface** for persistent AI sessions (agent-doc), and cross-language naming analysis (tagpath). A decade of **context engineering** - from a **domain ontology** system (existence-lang, 2014) through reactive frameworks to modern agentic systems. Deep expertise in event-driven architecture, **reactive programming**, and **collaborative intelligence** workflows where **agents and humans work in parallel**.

SKILLS

Context Engineering, **Spec-Driven Development**, **Agents**, **Rust**, **Python**, **TypeScript**, Zig, **React**, SvelteJS, **Docker**, **WebRTC**, REST, **GraphQL**, Node.js, Next.js, Django, FastAPI, Flask, Graphene, SQLAlchemy, Angular, Vue.js, Zod, Flutter, Go, Ruby on Rails, C#, **PostgreSQL**, TimescaleDB, Supabase, Firebase, **AWS**, **AWS** Lambda, **AWS** CDK, Grafana, Terraform, Google Cloud, Azure, CI/CD, Plesk, Heroku, Agentic Systems, Claude, Squarespace, WordPress, Wix

CAREER HIGHLIGHTS

- **Presence AI** - Architected real-time **multimodal** AI avatar pipeline - live video avatars, audio avatars, text avatars, and image generation on **NVIDIA H100 GPUs** via **WebRTC**
- **Open Source** - corky: git-native correspondence in Rust | agent-doc: **document-as-interface** for AI agents | RSpec: BDD core contributor | rr: pioneered proxy-based test doubles (now `patch` in Python)
- **Pivotal Labs** - Mentored developers across **20+ startups**. **Speaker at RailsConf 2008**. Early contributor to **Behavior Driven Development**
- **TrueCar / honk.com** - Led vehicle review platform development at honk.com (**acquired by TrueCar**); senior product team at TrueCar
- **Milyoni** - **Social Entertainment Platform** for **Warner Brothers**, Lions Gate, Paramount Studios, Universal Music Group
- **Censible** - 8-year ESG analytics platform - Martin Investments analyst portal, **ESG Portfolio Rebalancing** system
- **Context Engineering** - **Domain ontology** system (existence-lang) since 2014 - giving agents and humans a consistent vocabulary across code, correspondence, and data contexts
- **Reactive Frameworks** - 15-year lineage: backbone-signal → ctx-core → rmemo → relysjs - applied to modern agentic systems

AI PRODUCTS SHIPPED

- **Presence AI Avatar Pipeline** - Architected and shipped a production real-time **multimodal** AI pipeline: live video avatars (**facial landmark detection**, lip-sync inference, frame sequencing), audio avatars, text avatars, and image generation on **NVIDIA H100 GPUs** via **WebRTC**. Defined the **agentic development methodology** used by the engineering team.
- **corky** - Built and shipped a **git-native correspondence system** in Rust that syncs email, Telegram, and Slack into scoped markdown mailboxes with per-contact agent context and topic-based routing. Includes Whisper-

powered diarized transcription for voice messages, social media posting with scheduled delivery, and bidirectional sync across providers. Prototyped in Python, then **spec-driven** rewrite to Rust in under 1 day. Used daily for professional correspondence.

- **agent-doc** - Created and shipped a **document-as-interface** tool for **persistent AI conversations**: offline markdown editing, diff-based submissions, CRDT-based **streaming merge** for real-time human+agent co-editing without conflicts. Used as primary development interface.
- **agent-kit** - Built a Rust SDK for CLI tools integrating with AI **agent loops**: session management, tmux pane routing, instruction file discovery (AGENTS.md/CLAUDE.md), and snapshot-based state tracking. Powers agent-doc and corky's agent integrations.
- **tmux-router** - Declarative tmux pane routing in Rust: sync editor layouts to tmux sessions, multiplexed agent orchestration across concurrent development contexts.
- **tagpath** - Built and shipped a cross-language identifier analysis tool in Rust: convention detection, semantic search, alias generation, and tag co-occurrence graphs across 14 tree-sitter grammars. Published to crates.io.

EXPERIENCE

PRESENCE AI

April 2024 - February 2026 (1 year 10 months)

Lead Backend Engineer, AI Pipeline

- Full-stack feature development: Next.js + React frontend ↔ GraphQL (urql) ↔ API Services ↔ **Multimodal Avatar Agents** + PostgreSQL. TypeScript monorepo with 8 internal packages (@presence-ai/api, design-system, **livekit**, analytics, credits, monitoring, error, web)
- Architected and shipped the entire **real-time video pipeline**: **facial landmark detection**, **LatentSync lip-sync** inference (VAE decode, denoising, restoration), idle video generation, and frame sequencing on **NVIDIA H100 GPUs** via **WebRTC** using **LiveKit**
- Architected a **multi-service AI monorepo** with **closed-loop agentic workflows** and **self-auditing development infrastructure** (Claude Code / Codex)
- Developed an iterative **agentic development methodology**: lightweight living specs refined through **closed-loop agent auditing** and continuous **human-in-the-loop feedback**, accelerating design velocity while staying in flow with the emerging design
- Built reusable async primitives for high-quality **agent-authored code** and human comprehension: task lifecycle management, scoped cancellation, lazily resolved dependencies that decouple initialization order and cascade invalidation through the context, runtime profile logging, and stable cross-stack naming to maintain conceptual coherence for both **human and agent reasoning**
- Improved lip-sync quality through **landmark temporal smoothing**, LatentSync model upgrades, and optimized **Whisper audio feature extraction** via improved audio stream batching
- Set up Grafana-based resource monitoring for GPU compute infrastructure on RunPod

OPEN SOURCE

2006 - Present

*Creator / Solo Developer (Early contributor to **Behavior Driven Development**)*

CORKY - GIT-BASED CORRESPONDENCE TOOLKIT (RUST) GITHUB.COM/BTAKITA/CORKY | DOCS

- Built a **git-native correspondence system** that syncs topics, contacts, and threads from email (IMAP), Telegram, and Slack into scoped markdown mailboxes, with **per-topic and per-contact agent context** that propagates across collaborators via **3-way merge**
- Designed multi-scope mailbox architecture where each collaborator receives only relevant threads and context, with bidirectional sync using content hashing (FNV-1a) and git-based conflict resolution for **offline-first collaboration**
- Implemented **crash-safe incremental IMAP sync** with **streaming merge**, bulk import for Slack and Telegram archives, and label-based routing rules that fan out threads to multiple mailboxes
- Dog-fooding daily for personal and professional correspondence across Gmail and Protonmail accounts

AGENT-DOC - DOCUMENT-AS-INTERFACE FOR INTERACTIVE SESSIONS WITH AI AGENTS (RUST) GITHUB.COM/BTAKITA/AGENT-DOC

- Created a **document-as-interface** model for **persistent AI conversations**: edit a markdown file offline, submit diffs, agent responds inline - all tracked through git commits for full session history and branch-based workflows
- Engineered write-safe **concurrent editing** via CRDT-based **streaming merge**, enabling real-time human+agent co-editing without conflicts - evolved from **3-way merge** to character-level conflict-free resolution
- Built **agent-agnostic core** with **pluggable LLM backends**, automatic pre-commit of user changes, and **snapshot-based diffing** so only changed content is sent to the agent
- Dog-fooding as the primary interface for planning and iterating on both corky and agent-doc development

AGENT-KIT - SDK FOR AI AGENT CLI TOOLS (RUST) CRATES.IO/CRATES/AGENT-KIT

- Built a Rust SDK for CLI tools that integrate with AI **agent loops**: session management, tmux pane routing, instruction file discovery, and snapshot-based state tracking
- Powers agent-doc and corky's agent integrations - shared foundation for session lifecycle, config resolution, and concurrent write safety

TMUX-ROUTER - DECLARATIVE TMUX PANE ROUTING (RUST) CRATES.IO/CRATES/TMUX-ROUTER

- Declarative layout engine syncing editor state to tmux sessions - multiplexed agent orchestration across concurrent development contexts
- Auto-claims panes for agent sessions, routes documents to the correct editor, and maintains layout consistency across window resizes

OTHER CONTRIBUTIONS:

- *lazily-rs* - Creator. Reactive signals library in Rust (crates.io) with cross-language ports maintaining consistent semantics across Rust, Python, and Zig
- *lazily-py* - Creator. Lazy evaluation with **context caching** in Python
- *lazily-zig* - Creator. Cross-platform & thread-safe lazy evaluation with **context caching** in Zig
- *ctx-core* - Creator. Explicit, scalable cross-platform, **reactive contexts** written in JavaScript
- *rmemo (Reactive Memo)*/*RelementJS* - Creator. Smallest & most composable **reactive isomorphic UI library** in JavaScript
- *rappstack* - Creator. Full-stack framework using BunJS & Elysia for extensible web app modules
- *rebuildjs* - Creator. Simple hackable alternative to Vite for Multi-Page Apps
- *relysjs* - Creator. **Reactive web app server** for MPAs with middleware and build APIs
- *hyopjs* - Creator. Tiny hydration library (starting at 61 B)
- *Generic Query Analyzer* - Creator. .NET WinForms query analyzer for OleDB data sources including MS Access
- *SvelteJS* - Contributor (**breadth-first queueing algorithm**)
- *Nano Stores* - Contributor (**breadth-first queueing algorithm**)
- *rr (double Ruby)* - Creator. Test double framework for Ruby. Pioneered proxy-based test doubles (mock.proxy, stub.proxy) - pattern now standard as `patch` in Python's pytest/unittest
- *RSpec* - Core contributor to the Ruby **Behavior Driven Development** (BDD) testing framework
- *backbone-signal* - Creator. **Signal & Slots reactive API** for Backbone Models
- *Screw Unit* - Co-creator. BDD testing framework for JavaScript
- *existence-lang* - Creator. **Domain ontology** system defining scope, context, resolution, and abstraction - gives agents and humans a consistent vocabulary across code, correspondence, and data contexts (since 2014, formerly philosophy)
- *Desert* - Co-creator. Rails plugin framework with model/view/controller sharing

BRIAN TAKITA

January 2019 - February 2026 (7 years 1 month)

Full-Stack Engineer

Manage, develop, & ensure success of short-term freelance software projects. Uses subcontractors for development. Web and mobile Flutter apps across all industries.

- **BestInform** - European information platform for structured data, reports, and analytics. Pixel-perfect Figma implementation; real-time support messaging. Angular 14, PrimeNG, Angular Material, Firebase. bestinform.eu
- **Guard App** - Security service platform connecting guards, companies, and customers. Admin portal and mobile apps for patrol management, incident reporting, NFC verification. Flutter, Angular, Firebase, Java Spring Boot.
- **Drivemond** - Taxi and ride-hailing platform for Chicago. Passenger and driver mobile apps with real-time matching and GPS tracking. Flutter, AWS.
- **Galler AI** - AI-powered image generation and editing platform in the browser. React frontend with Figma design system; cropping, clipping, drawing tools. React, TypeScript.
- **SafeSchoolz (Aegis Platform)** - AI safety platform detecting firearms and alerting authorities. Built gun detection model (~85% mAP). Android, .NET, Python/Jupyter, Twilio.
- **Menu.com** - AWS CDK + Lambda serverless functions integrated with AWS Fargate Windows .NET API instances on ECS for food ordering platform. Docker, AWS ECS, IAM.
- **IIPM** - Membership and certification platform for Irish pension specialists. Member and admin portals. WordPress, PHP, JavaScript/jQuery, Stripe. iipm.ie

CENSIBLE

January 2016 - April 2024 (8 years 4 months)

Lead Product Developer

Developed web solutions for Financial Services, centered around an Environment, Social, Governance (ESG) lens. These tools helped investors align their investments with their values & manage exposure to ESG topics.

- Built Martin Investments analyst portal with ESG rating algorithms, XLSX export handling **10,000+ rows**, and SVG/PDF report generation
- Built ESG Portfolio Rebalancing system and ESG analytics platform - censible.co

SOCIALCHORUS

February 2018 - August 2018 (7 months)

Senior Software Engineer

- Product development on the Studio team
- Converted legacy Backbone frontend to Svelte & React
- ES5 to ES6/ES2017 migration
- Created best practices for front-end component development

NEO INNOVATION, INC.

September 2015 - December 2015 (4 months)

Software Development Consultant

Built a **document-as-interface** application for Tenfore Holdings - a research pipeline where executives email contacts and contextual notes, which are semantically matched to companies and contacts in Salesforce, then routed to the analyst team for refinement. Early exploration of the **document-as-interface** pattern later evolved into agent-doc (2024).

CRYSTALCOMMERCE

January 2015 - September 2015 (9 months)

Senior Software Developer

Migrated production Rails app. Re-architected the front end using Node.js, Browserify, Gulp, and Backbone.

RUNDAVOO

March 2013 - December 2014 (1 year 10 months)

Platform Architect / Lead Front End Developer

Led front-end development for Rundavoo, a **Google top 75 app** in 2014. Created a responsive, multi-platform (desktop, tablet, mobile, server) application that integrated with several APIs via CORS and a proxy server.

Technologies: HTML5, CSS3, Node.js, Browserify, Jasmine, **reactive programming**, Cloudflare, PhoneGap

MILYONI, INC.

March 2012 - March 2013 (1 year 1 month)

Architect / Lead Developer

Lead developer for **Social Entertainment Platform** including video, social marketing, gamification, **real-time social interaction**, and interactive engagements synced with video. Platform used by **Warner Brothers**, Lions Gate, Paramount Studios, Universal Music Group, Hearst Media, Starz, Ovation TV, Astro, Focus Features, Funimation, Sundance Now.

Mentored junior & mid-level developers. Proponent of flattening communication across the organization.

Technologies: Node.js, Backbone.js, Rails, Postgres, Redis, Heroku

TRUECAR / HONK.COM

August 2009 - February 2012 (2 years 7 months)

Lead Developer → Senior Software Engineer

Led development of honk.com, a vehicle review and white-label platform used by USAA and Wall Street Journal Autos. Honk had News Corp as an investor and was **acquired by TrueCar**. Partnered on product vision for social car shopping that was key to the acquisition.

At TrueCar, senior member of advanced product team - introduced new products, partner integrations (WSJ.com, USAA.com), and devops automation (Chef for EC2, workstations, datacenter servers). Rails 3 upgrade. Led initiatives to share agile knowledge across the organization.

- Full-stack Ruby on Rails + Sinatra development with **Lucene search**
- Led security review and compliance process for USAA partnership integration
- High Availability deployment on EC2 and Engine Yard
- JavaScript client/server **event framework** (jelly)
- Chef scripts for server and workstation automation

PIVOTAL LABS

January 2006 - August 2009 (3 years 7 months)

Agile Engineer

Agile Development in the XP tradition. TDD, Pair Programming, Daily Standups, Retrospectives. Mentored developers across **20+ startups**, bootstrapping their engineering process, team, and initial product. **Speaker at RailsConf 2008**. Leadership role in common code, practices, and open source.

Projects: Honk, Maven Link, KGB.com, Grockit, KDA Research, Real Girls Media, heavy.com, Bringo, CommunityWalk

Open source contributions:

- Erector - Ruby Builder and Widget framework for generating HTML
- JS Test Server - organized JavaScript TDD
- Unison - **event-driven relational algebra** library for Ruby

PEER TO PATENT PROJECT

2007

Consultant

Key contributor to launching the **Peer to Patent** pilot application, led by New York Law School. A tool to allow the crowd to review and submit prior art for pending patents.

NASPP February 2004 - January 2006 (1 year 11 months)

IT Analyst

- SQL Server DB Admin
- Developed internal CRM with Rails, .NET & MS Access
- Website ASP development
- Reports for Sales Trends and forecasting

FLEXTRONICS May 2001 - February 2002 (9 months)

Web Application Developer Intern

Developed corporate intranet using Microsoft technologies (ASP, SQL Server).

EDUCATION

University of the Pacific BS, **Engineering Physics** (1997 - 2002)

CERTIFICATIONS

- Advanced Permaculture Course in Teaching
- Permaculture Design Certificate

LANGUAGES

English: Native